

# Notes: Chevalier & Ellison (QJE, 1999)

## Career Concerns of Mutual Fund Managers

### Contents

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                     | <b>3</b>  |
| <b>2</b> | <b>Data and measurement</b>                                            | <b>3</b>  |
| 2.1      | Sample construction . . . . .                                          | 3         |
| 2.2      | Manager age and career stage . . . . .                                 | 4         |
| 2.3      | Performance: Jensen alpha . . . . .                                    | 4         |
| 2.4      | Risk and boldness variables . . . . .                                  | 4         |
| 2.5      | Termination and promotion outcomes . . . . .                           | 4         |
| <b>3</b> | <b>Models and empirical specifications</b>                             | <b>5</b>  |
| 3.1      | Basic termination-performance probit . . . . .                         | 5         |
| 3.2      | Semiparametric termination specification . . . . .                     | 5         |
| 3.3      | Direct effects of unconventional behavior . . . . .                    | 5         |
| 3.4      | Behavior regressions . . . . .                                         | 5         |
| 3.5      | Investor-flow specification . . . . .                                  | 6         |
| 3.6      | Promotion-performance model . . . . .                                  | 6         |
| <b>4</b> | <b>Identification and empirical design</b>                             | <b>6</b>  |
| 4.1      | What identifies career concerns . . . . .                              | 6         |
| 4.2      | Why age is central . . . . .                                           | 6         |
| 4.3      | What the paper cannot fully prove . . . . .                            | 7         |
| <b>5</b> | <b>Tables and figures</b>                                              | <b>7</b>  |
| 5.1      | Tables I–II: sample structure and career outcomes . . . . .            | 7         |
| 5.2      | Table III and Figure I: termination-performance relationship . . . . . | 8         |
| 5.3      | Tables IV–V: direct penalties for boldness . . . . .                   | 9         |
| 5.4      | Tables VI–VII: managerial behavior and investor flows . . . . .        | 10        |
| 5.5      | Table VIII, Figure II, and Table IX: promotion incentives . . . . .    | 11        |
| <b>6</b> | <b>Short summary</b>                                                   | <b>12</b> |
| <b>7</b> | <b>Conclusion</b>                                                      | <b>12</b> |
| 7.1      | Core conclusion . . . . .                                              | 12        |

|     |                              |    |
|-----|------------------------------|----|
| 7.2 | Economic intuition . . . . . | 13 |
| 7.3 | Takeaway . . . . .           | 13 |

# 1 Big picture

- **Question.** Do mutual fund managers' career concerns affect their portfolio choices?
- **Core setting.** The paper studies managers of growth and growth-and-income mutual funds during 1992–1994, using Morningstar data on manager identity, fund returns, assets, sector weights, and career outcomes.
- **Key career outcome.** A manager is treated as *terminated* if he loses the current fund and does not reappear managing a fund with greater total assets. This separates negative career outcomes from promotions or lateral moves.
- **Main labor-market result.** Termination is performance-sensitive: low Jensen alpha raises the probability of losing the job. The sensitivity is stronger for younger managers.
- **Shape of incentives.** For young managers, the termination-performance relationship is steep for negative alpha but flat for positive alpha. This creates a convex penalty from performance risk.
- **Risk-taking implication.** Young managers have an incentive to avoid unsystematic risk because bad idiosyncratic draws are punished while good idiosyncratic draws are not symmetrically rewarded.
- **Herding implication.** Young managers also face direct penalties for holding unconventional sector weights or unusual risk levels, especially after poor performance.
- **Behavioral evidence.** Younger managers hold less unsystematic risk and have more conventional portfolios than older managers.
- **Investor-flow evidence.** Investors appear to react to managerial changes, but the evidence is weaker than the termination and behavior results.
- **Promotion evidence.** Promotions are rare. Promotion incentives can encourage some boldness, especially after good performance, but they are smaller than termination incentives.
- **Economic takeaway.** Delegated portfolio management is shaped not only by fund-investor incentives but also by internal labor-market incentives. Career concerns can make young managers safer and more conventional.

## 2 Data and measurement

### 2.1 Sample construction

- The main sample contains managers who were sole managers of growth or growth-and-income mutual funds at the start of 1992, 1993, or 1994.
- Funds are followed through 1995, unless the fund disappears earlier.
- The termination and promotion regressions are fund-manager-year observations for 1992–1994.
- The behavior regressions use a larger 1992–1995 sample because they do not require observing the next year's career outcome.
- The flow regressions use a smaller sample after excluding very small funds, closing funds, and funds with extreme returns.

**Interpretation.** The design focuses on managers holding visible, high-level portfolio-manager positions. This is useful because performance is observable, actions are partly observable, and career outcomes can be tracked.

## 2.2 Manager age and career stage

- Manager age is the main proxy for career stage.
- Morningstar reports college and advanced-degree dates; the authors infer age by assuming college graduation at age 21.
- When birth year is directly available, it is used instead.
- Young managers are those below 45; older managers are those at least 45 in the nonparametric exercises.

**Interpretation.** Age proxies for how much the market has already learned about the manager. A younger manager’s ability is less known, so one performance signal can matter more for his career.

## 2.3 Performance: Jensen alpha

For each fund-year, the paper estimates a market model using monthly returns:

$$R_{i,m} - R_{f,m} = \alpha_{i,t} + \beta_{i,t}(R_{M,m} - R_{f,m}) + \varepsilon_{i,m}. \quad (1)$$

The performance variable is annualized Jensen alpha:

$$\text{Alpha}_{i,t} = \alpha_{i,t}. \quad (2)$$

**Interpretation.** Alpha is the risk-adjusted excess return that fund families and outside observers can use to infer managerial ability.

## 2.4 Risk and boldness variables

The paper constructs four main portfolio-choice variables:

$$\text{UnsysRisk}_{i,t} = \sqrt{\widehat{\text{Var}}(\varepsilon_{i,m})}, \quad (3)$$

$$\text{SectorDeviation}_{i,t} = \left[ \sum_s (w_{i,s,t} - \bar{w}_{g(i),s,t})^2 \right]^{1/2}, \quad (4)$$

$$\text{UnsysDeviation}_{i,t} = |\text{UnsysRisk}_{i,t} - \overline{\text{UnsysRisk}}_{g(i),t}|, \quad (5)$$

$$\text{BetaDeviation}_{i,t} = |\beta_{i,t} - \bar{\beta}_{g(i),t}|. \quad (6)$$

Here  $g(i)$  is the fund’s objective group, either growth or growth-and-income.

**Interpretation.** The first variable measures idiosyncratic risk. The last three measure how far a fund deviates from the herd in sector weights, idiosyncratic risk, and market beta.

## 2.5 Termination and promotion outcomes

- A separation occurs if the fund disappears or if the sole manager no longer manages the fund.
- A negative career outcome, called termination, is recorded when the manager disappears from fund management or reappears managing fewer total assets.
- A positive career outcome, called promotion, is recorded when the manager reappears managing greater total assets.

**Interpretation.** Tracking postseparation careers matters because not all manager departures are firings. Some are promotions.

### 3 Models and empirical specifications

#### 3.1 Basic termination-performance probit

The baseline latent-index specification is

$$\text{Termination}_{i,t}^* = \beta_0 + \beta_1 \text{Alpha}_{i,t} + \beta_2 \text{Alpha}_{i,t} (\text{MgrAge}_{i,t} - \overline{\text{Age}}) + \beta_3 \text{Alpha}_{i,t-1} + \beta_4 \text{Alpha}_{i,t-2} \quad (7)$$

$$+ \beta_5 \text{MgrAge}_{i,t} + \beta_6 \text{Age60+}_{i,t} + \beta_7 \text{GrowIncDummy}_{i,t} + \beta_8 \text{Year92}_t + \beta_9 \text{Year93}_t + \varepsilon_{i,t}, \quad (8)$$

$$\text{Termination}_{i,t} = \mathbf{1}\{\text{Termination}_{i,t}^* > 0\}. \quad (9)$$

**Interpretation.** The coefficient on current alpha is expected to be negative. The age interaction tests whether performance matters less as managers become older and more established.

#### 3.2 Semiparametric termination specification

To study the shape of the firing rule, the paper estimates

$$\text{Termination}_{i,t} = f(\text{Alpha}_{i,t}) + \beta_1 \text{Alpha}_{i,t-1} + \beta_2 \text{Alpha}_{i,t-2} + \beta_3 \text{MgrAge}_{i,t} + \beta_4 \text{GrowIncDummy}_{i,t} + \beta_5 \text{Age60+}_{i,t} + \beta_6 \text{Year92}_t + \beta_7 \text{Year93}_t \quad (10)$$

**Interpretation.** The nonparametric component  $f(\cdot)$  shows whether the implicit incentive scheme is linear, concave, or convex in performance.

#### 3.3 Direct effects of unconventional behavior

For each boldness measure  $B_{i,t} \in \{\text{SectorDeviation}, \text{UnsysDeviation}, \text{BetaDeviation}\}$ , the termination model includes

$$B_{i,t} \mathbf{1}\{\text{Alpha}_{i,t} \geq 0\}, \quad B_{i,t} \mathbf{1}\{\text{Alpha}_{i,t} < 0\}, \quad B_{i,t} (\text{MgrAge}_{i,t} - \overline{\text{Age}}) \mathbf{1}\{\text{Alpha}_{i,t} \geq 0\}, \quad B_{i,t} (\text{MgrAge}_{i,t} - \overline{\text{Age}}) \mathbf{1}\{\text{Alpha}_{i,t} < 0\}. \quad (11)$$

**Interpretation.** This asks whether being bold has a direct career effect after controlling for realized performance. The theory predicts especially strong penalties for failed boldness among young managers.

#### 3.4 Behavior regressions

To test whether managers respond to incentives, the paper estimates regressions such as

$$Y_{i,t} = \gamma_0 + \gamma_1 \text{MgrAge}_{i,t} + \gamma_2 \log(\text{Assets}_{i,t}) + \gamma_3 \text{GrowIncDummy}_{i,t} + \gamma_4 \text{Year92}_t + \gamma_5 \text{Year93}_t + \gamma_6 \text{Year94}_t + u_{i,t}, \quad (12)$$

where  $Y_{i,t}$  is UnsysRisk, SectorDeviation, UnsysDeviation, or BetaDeviation.

**Interpretation.** A positive age coefficient means older managers take more idiosyncratic risk or deviate more from the objective-group norm. That is the behavioral prediction from stronger career concerns for young managers.

### 3.5 Investor-flow specification

The paper estimates a nonlinear flow model:

$$\begin{aligned} \text{NetInflow}_{i,t+1} = & (1 + \kappa_0 \text{MChange}_{i,t}) \sum_k \gamma_k \text{FundAge}_{k,i,t} \text{Alpha}_{i,t} + \beta_1 \text{Alpha}_{i,t-1} + \beta_2 \text{Alpha}_{i,t-2} \\ & + \sum_k \delta_k \text{FundAge}_{k,i,t} + \lambda_0 + \lambda_1 \log(\text{Assets}_{i,t}) + \lambda_2 \text{Alpha}_{i,t+1} + \lambda_3 \text{Year92}_t + \lambda_4 \text{Year93}_t + \varepsilon_{i,t+1}, \end{aligned} \quad (13)$$

$$(14)$$

with

$$\text{NetInflow}_{i,t+1} = \frac{\text{Assets}_{i,t+1} - \text{Assets}_{i,t}}{\text{Assets}_{i,t}} - r_{i,t+1}. \quad (15)$$

**Interpretation.** The  $\kappa_0$  parameter measures whether a manager change makes flows less sensitive to past performance, as would occur if investors discount returns earned under the departed manager.

### 3.6 Promotion-performance model

The promotion analysis mirrors the termination analysis:

$$\text{Promotion}_{i,t}^* = \theta_0 + \theta_1 \text{Alpha}_{i,t} + \theta_2 \text{Alpha}_{i,t} (\text{MgrAge}_{i,t} - \overline{\text{Age}}) + \theta_3 \text{Alpha}_{i,t-1} + \theta_4 \text{Alpha}_{i,t-2} + \dots + \eta_{i,t}. \quad (16)$$

**Interpretation.** Promotions could counteract firing incentives by rewarding successful boldness. The paper finds some evidence of this, but promotion probabilities are low.

## 4 Identification and empirical design

### 4.1 What identifies career concerns

- The main variation is cross-sectional and time-series variation in manager performance, age, and portfolio choices.
- The paper compares how the performance-termination relationship differs for young and old managers.
- It then asks whether the behavior of young managers differs in exactly the direction implied by their stronger termination incentives.

**Interpretation.** The design does not randomly assign career concerns, but it links three pieces: termination rules, implied incentives, and observed behavior.

### 4.2 Why age is central

- Younger managers are less known to employers and the market.
- Learning models predict that a given performance signal should affect beliefs more for young managers.
- The estimates show stronger performance sensitivity and stronger penalties for unconventional failed behavior among young managers.

**Interpretation.** Age is both a proxy for reputation and a source of heterogeneity in implicit incentives.

### 4.3 What the paper cannot fully prove

- The paper does not observe explicit compensation contracts.
- Manager age is an imperfect proxy for career stage and information about ability.
- Portfolio choices and termination may both reflect unobserved fund-family strategies.
- The sample period is short and covers only growth and growth-and-income funds.
- The evidence is strongest for termination incentives and weaker for investor flows and promotions.

**Interpretation.** The paper is best read as evidence that career concerns plausibly shape mutual fund behavior, not as a randomized causal experiment.

## 5 Tables and figures

This section keeps adjacent tables and figures together to save space and make comparison faster. All visuals are directly cropped from the uploaded paper.

### 5.1 Tables I–II: sample structure and career outcomes

**Read:**

- Table I reports 1,320 fund-manager-year observations in the termination and promotion sample.
- The mean termination rate is 15.5 percent; the mean promotion rate is 2.8 percent.
- Mean manager age is about 44.2 years.
- The average fund has annualized Jensen alpha of about -0.62 percent and unsystematic risk of about 4.7 percent.
- Table II shows 242 total separations. Managers who disappear from the data have mean alpha of -2.5 percent, while managers who reappear managing greater assets have mean alpha of +1.5 percent.

**Economic message.** The tables justify the key outcome definitions. A separation is not always bad, so tracking postseparation assets is essential for distinguishing termination from promotion.

TABLE I  
SUMMARY STATISTICS

| Variable                                         | Number of observations | Mean    | Standard deviation |
|--------------------------------------------------|------------------------|---------|--------------------|
| Termination and promotion regressions: 1992-1994 |                        |         |                    |
| Termination                                      | 1320                   | 0.155   | 0.362              |
| Promotion                                        | 1320                   | 0.028   | 0.165              |
| $\text{Alpha}_t$                                 | 1320                   | -0.0062 | 0.070              |
| $\text{Alpha}_{t-1}$                             | 1320                   | 0.0018  | 0.075              |
| $\text{Alpha}_{t-2}$                             | 1320                   | -0.0041 | 0.066              |
| Manager Age                                      | 1320                   | 44.220  | 9.781              |
| $\text{Age}^{60+}$                               | 1320                   | 0.067   | 0.250              |
| GrowthIncDummy                                   | 1320                   | 0.361   | 0.480              |
| Beta                                             | 1320                   | 0.992   | 0.261              |
| UnsysRisk                                        | 1320                   | 0.047   | 0.028              |
| SectorDeviation                                  | 1320                   | 0.202   | 0.107              |
| UnsysDeviation                                   | 1320                   | 0.018   | 0.018              |
| BetaDeviation                                    | 1320                   | 0.182   | 0.170              |
| ExpenseRatio                                     | 1276                   | 0.013   | 0.009              |
| $\log(\text{Assets})$                            | 1278                   | 4.467   | 1.935              |
| $\log(\text{FamilySize})$                        | 1300                   | 7.057   | 2.510              |
| ManagerTenure                                    | 1319                   | 3.352   | 4.694              |
| Year92                                           | 1320                   | 0.277   | 0.447              |
| Year93                                           | 1320                   | 0.324   | 0.468              |
| Year94                                           | 1320                   | 0.399   | 0.490              |
| Sample for behavior regressions: 1992-1995       |                        |         |                    |
| UnsysRisk                                        | 1835                   | 0.047   | 0.027              |
| SectorDeviation                                  | 1490                   | 0.200   | 0.105              |
| UnsysDeviation                                   | 1835                   | 0.018   | 0.017              |
| BetaDeviation                                    | 1835                   | 0.192   | 0.177              |
| ManagerAge                                       | 1835                   | 44.403  | 9.702              |
| $\log(\text{Assets})$                            | 1835                   | 4.398   | 2.032              |
| Growth funds                                     |                        |         |                    |
| UnsysRisk                                        | 1181                   | 0.054   | 0.028              |
| Beta                                             | 1181                   | 1.053   | 0.285              |
| Growth and income funds                          |                        |         |                    |
| UnsysRisk                                        | 654                    | 0.035   | 0.021              |
| Beta                                             | 654                    | 0.912   | 0.219              |
| Sample for flow regression: 1992-1994            |                        |         |                    |
| $\text{Flow}_{t-1}$                              | 1056                   | 0.177   | 0.675              |
| MChange                                          | 1056                   | 0.149   | 0.356              |
| $\text{Alpha}_t$                                 | 1056                   | -0.0016 | 0.062              |
| $\text{Alpha}_{t-1}$                             | 1056                   | 0.0056  | 0.074              |
| $\text{Alpha}_{t-2}$                             | 1056                   | -0.0017 | 0.064              |
| $\text{Alpha}_{t-1}$                             | 1056                   | -0.022  | 0.088              |
| FundAge                                          | 1056                   | 13.504  | 16.180             |
| $\log(\text{Assets})$                            | 1056                   | 5.002   | 1.509              |

Summary statistics for all of the variables used in the analysis are provided. The upper panel provides summary statistics for the data sample used in Tables II through V, VII, and IX. The middle panel provides the summary statistics for the data sample used in Table VI. The lower panel provides the summary statistics for the data sample used in Table VII.

(a) Table I: summary statistics

TABLE II  
THE POSTSEPARATION CAREERS OF MUTUAL FUND MANAGERS

| Category                                                                                                     | Number | Mean adj. alpha |
|--------------------------------------------------------------------------------------------------------------|--------|-----------------|
| Manager-fund-years                                                                                           | 1320   |                 |
| Total separations                                                                                            | 242    |                 |
| Instances in which fund disappeared                                                                          | 72     | -0.017          |
| Instances in which fund survived but manager did not remain with fund                                        | 170    | -0.013          |
| Separations in which manager disappears from data                                                            | 98     | -0.025          |
| Separations in which manager reappears in data, managing fewer total assets                                  | 106    | -0.011          |
| Separations in which manager reappears in data, managing greater total assets                                | 38     | 0.015           |
| Separations in which manager disappears from data and manager is >60 years old                               | 4      | -0.001          |
| Fraction of total separations of managers $\geq 45$ in which manager disappears                              | 0.45   | -0.025          |
| Fraction of total separations of managers $\geq 45$ in which manager reappears managing fewer total assets   | 0.43   | -0.004          |
| Fraction of total separations of managers $\geq 45$ in which manager reappears managing greater total assets | 0.12   | 0.020           |
| Fraction of total separations of managers <45 in which manager disappears                                    | 0.36   | -0.031          |
| Fraction of total separations of managers <45 in which manager reappears managing fewer total assets         | 0.44   | -0.018          |
| Fraction of total separations of managers <45 in which manager reappears managing greater total assets       | 0.20   | 0.012           |

This table provides information on the postseparation careers of managers who were in charge of a growth or income fund in 1992, 1993, or 1994. A manager is defined to have "separated" from his position if he was the sole manager of a fund in year  $t$  but not in year  $t+1$ . A separated manager "disappears" if the Morningstar database does not list him as a manager of any fund in year  $t+1$ . He "reappears" if he does manage a fund in year  $t+1$ . The manager "reappears managing greater total assets" if the total assets managed by the manager in year  $t$  are smaller than the total assets managed by the manager in year  $t+1$  divided by one plus the growth rate of the mutual fund industry from year  $t$  to year  $t+1$ . The mean adjusted alpha given for each class of managers is the mean of the difference between Jensen's alpha for each fund in year  $t$  and the mean alpha of the fund's objective category in year  $t$ .

(b) Table II: postseparation careers

## 5.2 Table III and Figure I: termination-performance relationship

Read:

- Table III shows that current alpha strongly reduces the probability of termination. In column (1), the alpha coefficient is -3.225.
- The interaction  $\text{Alpha}_t \times (\text{MgrAge} - \bar{\text{Age}})$  is positive and significant. Because the alpha coefficient is negative, this means older managers have weaker performance sensitivity.
- Lagged alphas are also negative, indicating that past performance affects career outcomes.
- Figure I shows the shape: for young managers, termination probability falls sharply as alpha moves from negative to zero, then becomes much flatter for positive alpha.
- For older managers, the relationship is flatter throughout.

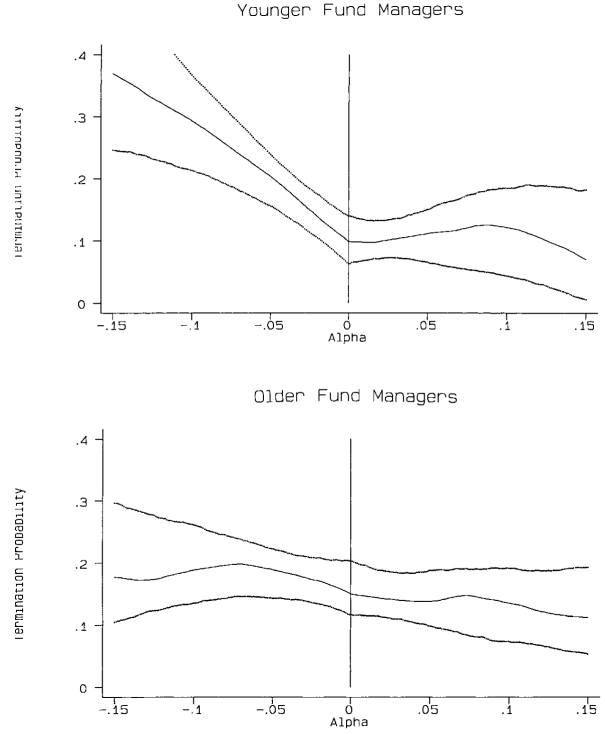
**Economic message.** Young managers face a strong downside penalty from bad performance but do not receive an equally strong career benefit from very good performance. That creates incentives to reduce idiosyncratic risk.

TABLE III  
BASIC TERMINATION-PERFORMANCE RELATIONSHIP

| Variable                                            | Dependent variable: Termination <sub>t</sub> |                   |                    |                   |                    |
|-----------------------------------------------------|----------------------------------------------|-------------------|--------------------|-------------------|--------------------|
|                                                     | (1)                                          | (2)               | (3)                | (4)               | (5)                |
| Alpha <sub>t</sub>                                  | -3.225<br>(0.734)                            | -3.842<br>(0.807) | -3.885<br>(0.793)  | -3.296<br>(0.768) | -3.198<br>(0.746)  |
| Alpha <sub>t</sub> × (MgrAge - $\bar{\text{Age}}$ ) | 0.259<br>(0.064)                             | 0.280<br>(0.077)  | 0.302<br>(0.070)   | 0.269<br>(0.070)  | 0.262<br>(0.079)   |
| Alpha <sub>t-1</sub>                                | -1.462<br>(0.673)                            | -1.346<br>(0.702) | -1.486<br>(0.691)  | -1.910<br>(0.705) | -1.405<br>(0.687)  |
| Alpha <sub>t-2</sub>                                | -0.753<br>(0.768)                            | -1.411<br>(0.872) | -1.220<br>(0.818)  | -1.194<br>(0.783) | -0.800<br>(0.783)  |
| ManagerAge                                          | -0.003<br>(0.006)                            | 0.0003<br>(0.006) | -0.0006<br>(0.006) | -0.005<br>(0.006) | -0.0010<br>(0.006) |
| Age60+                                              | 0.240<br>(0.190)                             | 0.229<br>(0.197)  | 0.211<br>(0.200)   | 0.314<br>(0.193)  | 0.293<br>(0.191)   |
| GrowthDummy                                         | 0.231<br>(0.089)                             | 0.248<br>(0.091)  | 0.251<br>(0.092)   | 0.242<br>(0.091)  | 0.236<br>(0.088)   |
| ExpenseRatio                                        |                                              | -9.284<br>(5.178) |                    |                   |                    |
| log (Assets)                                        |                                              |                   | 0.052<br>(0.024)   |                   |                    |
| log (Assets) × Alpha <sub>t</sub>                   |                                              |                   | 0.243<br>(0.367)   |                   |                    |
| log (FamilySize)                                    |                                              |                   |                    | 0.045<br>(0.018)  |                    |
| log (FamilySize) × Alpha <sub>t</sub>               |                                              |                   |                    | -0.270<br>(0.274) |                    |
| ManagerTenure                                       |                                              |                   |                    |                   | -0.015<br>(0.010)  |
| Alpha <sub>t</sub> × (MgrTen - $\bar{\text{Ten}}$ ) |                                              |                   |                    |                   | 0.026<br>(0.122)   |
| Year92                                              | -0.114<br>(0.106)                            | -0.101<br>(0.109) | -0.081<br>(0.110)  | -0.245<br>(0.112) | -0.106<br>(0.107)  |
| Year93                                              | -0.159<br>(0.105)                            | -0.287<br>(0.112) | -0.303<br>(0.113)  | -0.155<br>(0.106) | -0.168<br>(0.106)  |
| Constant                                            | -0.979<br>(0.254)                            | -1.009<br>(0.269) | -1.329<br>(0.290)  | -1.221<br>(0.305) | -1.021<br>(0.259)  |
| Number of observations                              | 1320                                         | 1276              | 1278               | 1300              | 1319               |

Each column is a probit specification in which the dependent variable takes the value of one if the manager was "terminated" from his position and zero otherwise. Each observation is a fund manager-year. Standard errors, adjusted for intrafund correlation of the errors, are in parentheses.

(a) Table III: basic termination-performance relationship



(b) Figure I: shape of termination-performance relationship

### 5.3 Tables IV–V: direct penalties for boldness

Read:

- Table IV estimates whether bold portfolio choices directly predict termination after controlling for performance.
- Sector deviation is strongly punished when alpha is negative: the coefficient on SectorDeviation(Alpha < 0) is 1.129.
- Unsystematic-risk deviation is also punished when alpha is negative: the coefficient on UnsysDeviation(Alpha < 0) is 4.917.
- The age interactions are generally negative in the failed-boldness cases, indicating that younger managers face stronger penalties.
- Table V converts the coefficients into predicted probability changes. A one-standard-deviation sector deviation raises termination probability by 4.5 percentage points for a 35-year-old with negative alpha.
- A one-standard-deviation unsystematic-risk deviation raises termination probability by 5.5 percentage points for a 35-year-old with negative alpha.

**Economic message.** Career concerns generate a herding motive. Young managers who fail while looking different are more likely to lose their jobs than managers who fail conventionally.

TABLE IV  
DIRECT EFFECTS OF DEVIATIONS FROM BEHAVIORAL NORMS

| Dependent variable: Termination <sub>t</sub>                    |                   |                   |                   |
|-----------------------------------------------------------------|-------------------|-------------------|-------------------|
| Independent variable                                            | (1)               | (2)               | (3)               |
| Alpha <sub>t</sub>                                              | -2.195<br>(0.927) | -2.142<br>(0.879) | -2.796<br>(0.876) |
| Alpha <sub>t</sub> × (MgrAge - $\overline{\text{Age}}$ )        | 0.258<br>(0.098)  | 0.220<br>(0.092)  | 0.144<br>(0.093)  |
| Alpha <sub>t-1</sub>                                            | -1.606<br>(0.655) | -1.488<br>(0.700) | -1.474<br>(0.687) |
| Alpha <sub>t-2</sub>                                            | -0.888<br>(0.751) | -0.960<br>(0.807) | -0.845<br>(0.803) |
| ManagerAge                                                      | 0.007<br>(0.007)  | 0.013<br>(0.007)  | 0.006<br>(0.007)  |
| Age60+                                                          | 0.347<br>(0.193)  | 0.253<br>(0.189)  | 0.250<br>(0.192)  |
| GrowIncDummy                                                    | 0.265<br>(0.090)  | 0.264<br>(0.092)  | 0.227<br>(0.091)  |
| SectorDev(Alpha ≥ 0)                                            | 0.169<br>(0.525)  |                   |                   |
| SectorDev(Alpha < 0)                                            | 1.129<br>(0.423)  |                   |                   |
| SectorDev × (MgrAge - $\overline{\text{Age}}$ )(Alpha ≥ 0)      | -0.069<br>(0.037) |                   |                   |
| SectorDev × (MgrAge - $\overline{\text{Age}}$ )(Alpha < 0)      | -0.077<br>(0.038) |                   |                   |
| UnsysDeviation(Alpha ≥ 0)                                       |                   | -5.456<br>(4.371) |                   |
| UnsysDeviation(Alpha < 0)                                       |                   | 4.917<br>(3.262)  |                   |
| UnsysDeviation × (MgrAge - $\overline{\text{Age}}$ )(Alpha ≥ 0) |                   | -0.792<br>(0.401) |                   |
| UnsysDeviation × (MgrAge - $\overline{\text{Age}}$ )(Alpha < 0) |                   | -0.889<br>(0.299) |                   |
| BetaDeviation(Alpha ≥ 0)                                        |                   |                   | -0.512<br>(0.434) |
| BetaDeviation(Alpha < 0)                                        |                   |                   | -0.089<br>(0.363) |
| BetaDeviation × (MgrAge - $\overline{\text{Age}}$ )(Alpha ≥ 0)  |                   |                   | -0.008<br>(0.038) |
| BetaDeviation × (MgrAge - $\overline{\text{Age}}$ )(Alpha < 0)  |                   |                   | -0.093<br>(0.041) |
| Year92                                                          | -0.150<br>(0.108) | -0.146<br>(0.111) | -0.101<br>(0.107) |
| Year93                                                          | -0.171<br>(0.106) | -0.172<br>(0.106) | -0.143<br>(0.107) |
| Constant                                                        | 1.531<br>(0.343)  | 1.664<br>(0.336)  | 1.337<br>(0.330)  |

Each column is a probit specification in which the dependent variable takes the value of one if the manager was "terminated" from his position and zero otherwise. Each observation is a fund manager-year. Standard errors, adjusted for intrafund correlation of the errors, are in parentheses.

(a) Table IV: direct effects of deviations from behavioral norms

TABLE V  
INCREASE IN TERMINATION PROBABILITY FROM ONE-STANDARD-DEVIATION  
INCREASE IN BOLDNESS

| Manager age<br>and performance | Measure of boldness |                |               |
|--------------------------------|---------------------|----------------|---------------|
|                                | SectorDeviation     | UnsysDeviation | BetaDeviation |
| Age 35 Alpha <sub>t</sub> < 0  | 0.045**             | 0.055**        | 0.032         |
| Age 35 Alpha <sub>t</sub> > 0  | 0.020               | 0.010          | -0.016        |
| Age 45 Alpha <sub>t</sub> < 0  | 0.027**             | 0.020          | -0.003        |
| Age 45 Alpha <sub>t</sub> > 0  | 0.004               | -0.022         | -0.019        |

This table uses the estimates from the previous table to construct predicted increases in the probability of termination implied by a one-standard-deviation increase in boldness for managers with different characteristics. An asterisk denotes significance at the 5 percent level. Two asterisks denotes significance at the 1 percent level.

(b) Table V: termination probability from boldness

## 5.4 Tables VI–VII: managerial behavior and investor flows

Read:

- Table VI tests whether managers behave in ways predicted by the career-concern incentives.
- Manager age positively predicts unsystematic risk, sector deviation, unsystematic-risk deviation, and beta deviation.
- Since young managers face stronger downside career risk, this means young managers hold safer and more conventional portfolios.
- Table VII studies flows. The estimated MChange parameter is around -0.24, suggesting that flows are less sensitive to past performance after a managerial change.
- The flow evidence is suggestive but weaker than the termination and behavior evidence.

**Economic message.** The behavior results are the central response evidence: young managers act as if they are trying to avoid the risks and deviations that increase termination probability.

TABLE VI  
PATTERNS IN MANAGERIAL BEHAVIOR

| Independent variables | Dependent variable   |                     |                      |                   |
|-----------------------|----------------------|---------------------|----------------------|-------------------|
|                       | UnsysRisk            | SectorDeviation     | UnsysDeviation       | BetaDeviation     |
| Constant              | 0.0478<br>(0.0054)   | 0.1400<br>(0.0290)  | 0.0165<br>(0.0040)   | 0.178<br>(0.027)  |
| ManagerAge            | 0.00035<br>(0.00012) | 0.0021<br>(0.0006)  | 0.00022<br>(0.00010) | 0.002<br>(0.0006) |
| log (Assets)          | -0.0021<br>(0.0004)  | -0.0074<br>(0.0019) | -0.0010<br>(0.0003)  | -0.008<br>(0.002) |
| GrowthIncDummy        | -0.0018<br>(0.0017)  | -0.0358<br>(0.0092) | -0.0072<br>(0.0012)  | -0.053<br>(0.010) |
| Year92                | 0.0076<br>(0.0018)   | 0.0029<br>(0.0058)  | 0.0034<br>(0.0013)   | -0.024<br>(0.011) |
| Year93                | 0.0028<br>(0.0012)   | 0.0016<br>(0.0055)  | -0.0014<br>(0.0009)  | 0.014<br>(0.012)  |
| Year94                | -0.0078<br>(0.0010)  | 0.0083<br>(0.0047)  | -0.0046<br>(0.0007)  | -0.074<br>(0.09)  |
| NOBS                  | 1835                 | 1835                | 1835                 | 1835              |
| R <sup>2</sup>        | 0.18                 | 0.08                | 0.10                 | 0.10              |

Each column is a regression specification in which some characteristic of the manager's portfolio choices is regressed on fund and manager characteristics. The first column has the fund's unsystematic risk level as the dependent variable; the second through fourth columns have our "herdiness" measures as the dependent

(a) Table VI: patterns in managerial behavior

TABLE VII  
DETERMINANTS OF MUTUAL FUND ASSET FLOWS

| Parameter      | Independent variables                           | Dependent variable: NetInflow <sub>t+1</sub><br>Estimation |                   |                   |                   |
|----------------|-------------------------------------------------|------------------------------------------------------------|-------------------|-------------------|-------------------|
|                |                                                 | NLS                                                        | NLS               | NL2SLS            | NL2SLS            |
| $\kappa_0$     | MChange <sub>t</sub>                            | -0.243<br>(0.185)                                          |                   | -0.242<br>(0.192) |                   |
| $\kappa_1$     | MChange <sub>t</sub> × (Alpha <sub>t</sub> < 0) |                                                            | -0.449<br>(0.209) |                   | -0.404<br>(0.261) |
| $\kappa_2$     | MChange <sub>t</sub> × (Alpha <sub>t</sub> > 0) |                                                            | 0.409<br>(0.382)  |                   | 0.427<br>(0.425)  |
| $\gamma_{01}$  | FundAge01 × Alpha <sub>t</sub>                  | 7.144<br>(0.811)                                           | 6.883<br>(0.801)  | 7.146<br>(0.801)  | 7.190<br>(0.799)  |
| $\gamma_{23}$  | FundAge23 × Alpha <sub>t</sub>                  | 3.299<br>(0.810)                                           | 3.378<br>(0.816)  | 3.310<br>(0.804)  | 3.689<br>(0.813)  |
| $\gamma_{46}$  | FundAge46 × Alpha <sub>t</sub>                  | 6.129<br>(0.742)                                           | 6.151<br>(0.744)  | 6.139<br>(0.737)  | 6.397<br>(0.743)  |
| $\gamma_{79}$  | FundAge79 × Alpha <sub>t</sub>                  | 3.345<br>(0.783)                                           | 3.399<br>(0.778)  | 3.352<br>(0.780)  | 3.735<br>(0.775)  |
| $\gamma_{10+}$ | FundAge10+ × Alpha <sub>t</sub>                 | 2.067<br>(0.490)                                           | 2.113<br>(0.476)  | 2.073<br>(0.486)  | 2.554<br>(0.476)  |
| $\beta_1$      | Alpha <sub>t-1</sub>                            | 1.100<br>(0.267)                                           | 1.167<br>(0.259)  | 1.105<br>(0.261)  | 1.151<br>(0.259)  |
| $\beta_2$      | Alpha <sub>t-2</sub>                            | 1.672<br>(0.304)                                           | 1.647<br>(0.299)  | 1.677<br>(0.300)  | 1.613<br>(0.298)  |
| $\delta_{01}$  | FundAge01                                       | 0.503<br>(0.059)                                           | 0.492<br>(0.059)  | 0.504<br>(0.059)  | 0.488<br>(0.059)  |
| $\delta_{23}$  | FundAge23                                       | 0.164<br>(0.054)                                           | 0.162<br>(0.054)  | 0.164<br>(0.054)  | 0.165<br>(0.054)  |
| $\delta_{46}$  | FundAge46                                       | 0.221<br>(0.054)                                           | 0.215<br>(0.054)  | 0.222<br>(0.054)  | 0.216<br>(0.054)  |
| $\delta_{79}$  | FundAge79                                       | 0.051<br>(0.057)                                           | 0.044<br>(0.057)  | 0.051<br>(0.057)  | 0.044<br>(0.056)  |
| $\lambda_0$    | Constant                                        | 0.257<br>(0.085)                                           | 0.256<br>(0.085)  | 0.255<br>(0.084)  | 0.260<br>(0.084)  |
| $\lambda_1$    | log (Assets <sub>t</sub> )                      | -0.030<br>(0.013)                                          | -0.031<br>(0.013) | -0.030<br>(0.013) | -0.030<br>(0.013) |
| $\lambda_2$    | Alpha <sub>t+1</sub>                            | 0.914<br>(0.231)                                           | 0.899<br>(0.230)  | 0.915<br>(0.229)  | 0.918<br>(0.229)  |
| $\lambda_3$    | Year92                                          | 0.014<br>(0.049)                                           | 0.016<br>(0.049)  | 0.015<br>(0.049)  | 0.015<br>(0.049)  |
| $\lambda_4$    | Year93                                          | -0.180<br>(0.045)                                          | -0.175<br>(0.045) | -0.179<br>(0.045) | -0.166<br>(0.045) |

This table presents nonlinear least squares and nonlinear two-stage least squares specifications. The 1056 observations are fund-years for growth and income funds from 1992–1994. To be included in the specification, a fund had to have assets of greater than \$10 million. The dependent variable, NetInflow<sub>t+1</sub>, is the growth rate of fund assets from the end of year t to the end of year t + 1 minus the fund's return in year t + 1. The nonlinearity stems from allowing the slope of the flow-performance relationship to differ between those funds whose manager changed between year t and year t + 1 and those funds whose managers did not change. Standard errors are in parentheses.

(b) Table VII: determinants of mutual fund asset flows

## 5.5 Table VIII, Figure II, and Table IX: promotion incentives

Read:

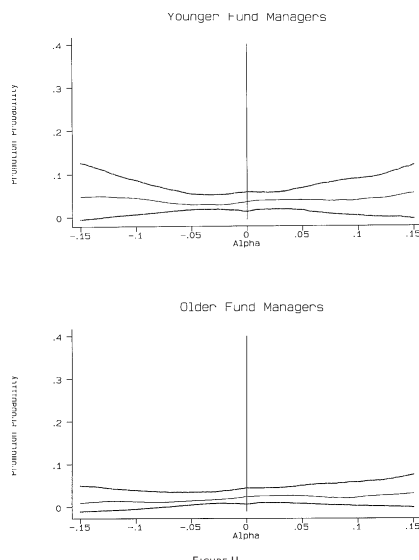
- Table VIII shows that promotion is rare and only weakly tied to performance.
- Manager age has a negative coefficient in the promotion probit, implying younger managers are more likely to be promoted.
- Figure II shows relatively flat promotion-performance curves, especially compared with the termination curves.
- Table IX shows that successful bold sector bets can raise promotion probability: for a 35-year-old with positive alpha, a one-standard-deviation sector deviation raises promotion probability by 1.9 percentage points.
- These promotion incentives can partly offset herding incentives, but they are smaller than the termination penalties.

**Economic message.** The strongest labor-market force is avoiding termination, not chasing promotion. Promotions provide some upside for boldness, but not enough to overturn the main conclusion.

TABLE VIII  
PROMOTION-PERFORMANCE RELATIONSHIP

| Independent variable                               | Dependent variable<br>Promotion <sub>it</sub> |
|----------------------------------------------------|-----------------------------------------------|
| Alpha <sub>it</sub>                                | 1.460<br>(1.233)                              |
| Alpha <sub>it</sub> × (MgrAge <sub>it</sub> − Age) | 0.010<br>(0.107)                              |
| Alpha <sub>it-1</sub>                              | 1.561<br>(0.893)                              |
| Alpha <sub>it-2</sub>                              | −0.438<br>(1.108)                             |
| ManagerAge                                         | −0.021<br>(0.009)                             |
| Age60+                                             | 0.429<br>(0.378)                              |
| GrowthDummy                                        | −0.054<br>(0.142)                             |
| Year92                                             | −0.297<br>(0.155)                             |
| Year93                                             | −0.835<br>(0.257)                             |
| Constant                                           | −0.771<br>(0.363)                             |
| Number of observations                             | 1320                                          |

(a) Table VIII: promotion-performance relationship



(b) Figure II: promotion-performance shape

TABLE IX  
INCREASE IN PROMOTION PROBABILITY FROM ONE-STANDARD-DEVIATION INCREASE IN BOLDNESS

| Manager age and performance    | Measure of boldness |                |               |
|--------------------------------|---------------------|----------------|---------------|
|                                | SectorDeviation     | UnsysDeviation | BetaDeviation |
| Age 35 Alpha <sub>it</sub> < 0 | −0.001              | −0.016         | −0.007        |
| Age 35 Alpha <sub>it</sub> > 0 | 0.019**             | 0.002          | 0.011         |
| Age 45 Alpha <sub>it</sub> < 0 | 0.002               | −0.008         | −0.001        |
| Age 45 Alpha <sub>it</sub> > 0 | 0.009*              | 0.007*         | 0.008         |

(c) Table IX: promotion probability from boldness

## 6 Short summary

- The paper studies whether mutual fund managers' career concerns affect their investment decisions.
- The sample consists of managers of growth and growth-and-income funds during 1992–1994, with some behavior tests through 1995.
- Termination is defined as losing the fund and not reappearing in charge of greater total assets.
- Termination is strongly negatively related to current and lagged Jensen alpha.
- Performance sensitivity is stronger for younger managers.
- For young managers, the firing rule is steep for negative alpha and flat for positive alpha.
- This makes unsystematic risk unattractive for young managers.
- Young managers are also penalized for deviating from sector and risk norms, especially after poor performance.
- The behavior regressions show that young managers hold less unsystematic risk and deviate less from the herd.
- Investor-flow reactions to manager changes are suggestive but weaker.
- Promotions are rare and create smaller countervailing incentives for successful boldness.
- The paper shows that internal labor-market incentives matter for delegated portfolio management.

## 7 Conclusion

### 7.1 Core conclusion

Chevalier and Ellison show that mutual fund managers face implicit career incentives that differ by age. Younger managers are punished more severely for poor performance and for failed unconventional choices.

These incentives help explain why younger managers take less unsystematic risk and stay closer to the portfolio choices of their peers.

## **7.2 Economic intuition**

A young manager's ability is uncertain. A bad return or a failed unconventional strategy is therefore interpreted as a stronger negative signal about ability. Because the labor market punishes the downside more than it rewards the upside, young managers have incentives to avoid idiosyncratic risk and to herd toward conventional portfolios.

## **7.3 Takeaway**

The paper expands the study of mutual fund incentives beyond the relationship between investors and fund companies. The manager's internal labor market matters too. Career concerns can make managers conservative, conventional, and less willing to take bold active bets, especially early in their careers.